Comparative cardiovascular effects of thiazolidinediones: systematic review and meta-analysis of observational studies

Objective To determine the comparative effects of the thiazolidinediones (rosiglitazone and pioglitazone) on myocardial infarction, congestive heart failure, and mortality in patients with type 2 diabetes. Design Systematic review and meta-analysis of observational studies. Data sources Searches of Medline and Embase in September 2010. Study selection Observational studies that directly compared the risk of cardiovascular outcomes for rosiglitazone and pioglitazone among patients with type 2 diabetes mellitus were included. Data extraction Random effects meta-analysis (inverse variance method) was used to calculate the odds ratios for cardiovascular outcomes with thiazolidinedione use. The I2 statistic was used to assess statistical heterogeneity. Results Cardiovascular outcomes from 16 observational studies (4 case-control studies and 12 retrospective cohort studies), including 810 000 thiazolidinedione users, were evaluated after a detailed review of 189 citations. Compared with pioglitazone, use of rosiglitazone was associated with a statistically significant increase in the odds of myocardial infarction (n=15 studies; odds ratio 1.16, 95% confidence interval 1.07 to 1.24; P<0.001; I2=46%), congestive heart failure (n=8; 1.22, 1.14 to 1.31; P<0.001; I2=37%), and death (n=8; 1.14, 1.09 to 1.20; P<0.001; I2=0%). Numbers needed to treat to harm (NNH), depending on the population at risk, suggest 170 excess myocardial infarctions, 649 excess cases of heart failure, and 431 excess deaths for every 100 000 patients who receive rosiglitazone rather than pioglitazone. Conclusion Among patients with type 2 diabetes, use of rosiglitazone is associated with significantly higher odds of congestive heart failure, myocardial infarction, and death relative to pioglitazone in real world settings.